

Teaching a Fracture Classifier Where to Look: CNN vs ScatNet, Six XAI Methods and a Joint Classifier–Detector on FracAtlas

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Abstract—We classify X-rays of FracAtlas (4,083 images, one in six fractured, with the radiologists’ fracture boxes) and ask not only *whether* a model is right but *where it looks*. (i) A CNN and a wavelet scattering network (ScatNet) that share the same classifier both stay at the level of always answering “not fractured” (80.9% and 80.0% test accuracy, F1 23–32%), while a pretrained ResNet18 reaches 91.5% (F1 73.7%). (ii) Six XAI methods, one of them (Occlusion) re-implemented from scratch and identical to Captum up to $9 \cdot 10^{-6}$, are scored for faithfulness (deletion test) and for pointing at the fracture: on ResNet18 the best method, Integrated Gradients, puts its hottest point inside the radiologist’s box only 23% of the time. (iii) We propose a *joint model*: one ResNet18 at 448 px with the same classifier and a built-in detector, trained together with three explanation losses (Grad-CAM inside the box; Grad-CAM, detector and mirrored X-ray agree; the answer changes when the fracture is blurred). It keeps 93.0% accuracy, doubles the Grad-CAM hit rate (35%→67%, $p < 0.001$), and its decision now depends on the fracture: blurring it drops $p(\text{fractured})$ from 84% to 3%. (iv) YOLOv8 remains the best box finder (79%); letting our model re-rank YOLO’s boxes does not place them better, but removes false boxes on healthy X-rays (AP@0.5 on all X-rays 40.7%→46.9%). A final pipeline decides 76% of the X-rays with 97% accuracy and misses 6% of the fractures.

Index Terms—fracture detection, scattering networks, explainable AI, pointing game, explanation-guided learning, object detection.

I. INTRODUCTION

A fracture classifier can be accurate for the wrong reasons: a metal plate, a cast or the body part can predict the label better than a thin fracture line. Explainable AI (XAI) methods draw a heatmap of the evidence, but a heatmap is only useful if it is *faithful* (it shows what the model uses) and if the model uses the *fracture*. We study both on FracAtlas [1], whose radiologists’ boxes let us measure where a model looks.

Our pipeline (Fig. 1) has five linked stages, each answering one question:

- **S1 Data.** How do we split without leakage? Near-duplicate X-rays are kept in the same split.
- **S2 Models.** CNN or ScatNet, with the same classifier? (ResNet18 as a reference.)

- **S3 Explanations.** Which of six XAI methods is faithful, and do the maps point at the fracture?
- **S4 Our method.** Can we *train* the model to look at the fracture, without losing accuracy?
- **S5 Detection.** Can our model help YOLOv8 [22] place the box, as in the pipeline of Linda [2]?

Main contributions.

- A **fair protocol**: a grouped split that keeps copies of an X-ray together, the same classifier for every model, McNemar tests, and a pointing-game hit rate against the radiologists’ boxes.
- **Six XAI methods compared** by faithfulness and by localisation, with Occlusion written from scratch.
- A **joint classifier–detector** trained with explanation losses, which doubles the Grad-CAM hit rate and makes the decision depend on the fracture region (blur test).
- An honest test of “**YOLO proposes, the classifier decides**”: it does not beat YOLO at placing boxes, but it removes YOLO’s false boxes on healthy X-rays.

II. RELATED WORK

Scattering networks. A scattering transform [3] cascades fixed wavelet filters, moduli and averaging. It is stable to small deformations and needs no training of its filters; we use the Kymatio implementation [4]. **XAI methods.** Gradient methods explain with the derivative of the score (Saliency [6], Guided Backprop [7], Integrated Gradients [8], Grad-CAM on the last convolutional layer [9]); perturbation methods remove parts of the input (Occlusion [10], LIME [11]). Faithfulness is measured by deleting the most important regions first [12]; localisation by the pointing game [13]. We use Captum [14]. **Explanation-guided learning.** Models can be trained so that their explanations fall on annotated regions: input gradients [15], attention maps with attention mining [16], or boxes with an Energy loss [17]. Attention maps of two views of an image can be made to agree [18], an idea close to contrastive learning with two views [19]. Heo *et al.* [20] warn that maps can be changed without changing the reasoning,

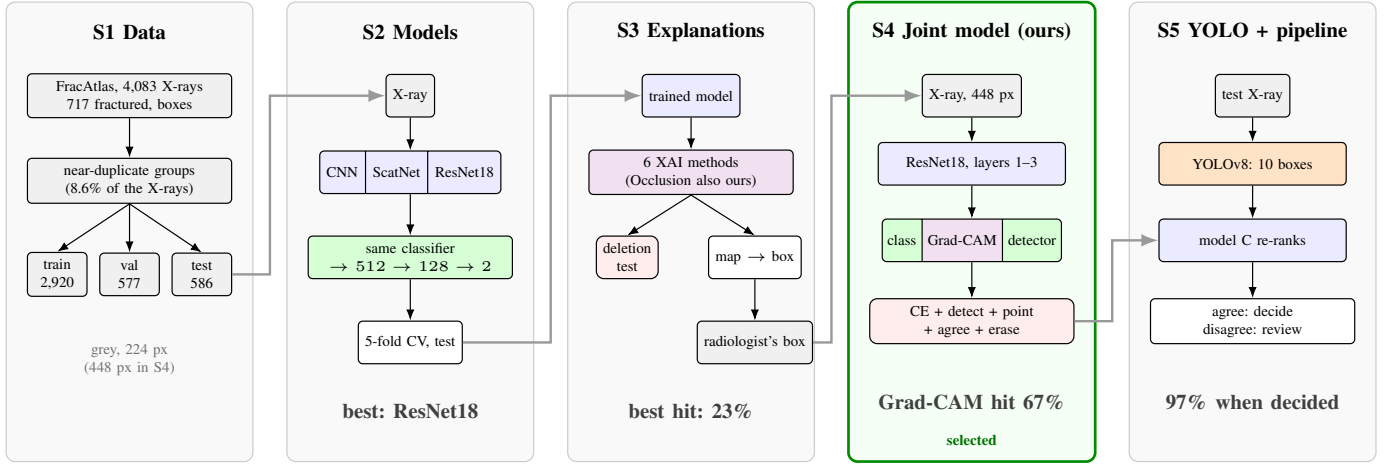


Fig. 1. **The pipeline.** One column per stage, read top to bottom; the grey arrows hand each result to the next stage. S1: a split that keeps copies of an X-ray together. S2: three feature extractors with the same classifier. S3: six XAI methods, scored by the deletion test and by whether their box lands on the radiologist’s box. S4 (green, our method): one network that classifies, detects and is taught where to look. S5: YOLO proposes boxes, our model picks one, and the two models act as two opinions. Grey: data; blue: trained network; green: heads; violet: XAI; red: losses; orange: YOLO.

TABLE I
THE GROUPED SPLIT (COPIES OF AN X-RAY NEVER CROSS SPLITS).

Split	X-rays	Fractured	% fractured	Boxes
Train	2,920	511	17.5	651
Val	577	102	17.7	136
Test	586	104	17.7	135

which motivates our blur test. **Detection.** YOLOv8 [22] is the reference detector on FracAtlas [1]; CenterNet [21] detects objects as the peaks of a centre heatmap, the design of our built-in detector. Linda [2] combines a classifier, YOLOv8 and XAI into a report; our S5 follows this idea on X-rays only.

III. DATA, NOTATION AND PROTOCOL

Data. FracAtlas has 4,083 X-rays of hands, legs, hips and shoulders; 717 are fractured (17.6%) and carry 922 radiologists’ boxes. Images are loaded in grey, resized to 224×224 (448 in S4) and scaled to $[-1, 1]$; black (-1) is the “removed” value of every XAI method. Label 0 (fractured) is the positive class.

No leakage (S1). Some X-rays appear more than once (flipped, resized or brighter). We compare 32×32 thumbnails under the 8 flips and rotations: 351 X-rays (8.6%) have a copy, in 158 groups. A stratified, grouped split keeps every group in one split (Table I); the cross-validation folds are grouped the same way.

Metrics. One X-ray in six is fractured, so always answering “not fractured” already gives 82.3% test accuracy. We therefore report F1, *fractures found* (recall), specificity and AUC, a 95% bootstrap interval of the accuracy, and the exact McNemar test between two models on the same X-rays. For explanations we report:

- **Deletion AUC:** remove the most important 16×16 patches first (set to black) and average the class prob-

TABLE II
SETTINGS (SHARED BY EVERY METHOD COMPARED IN A STAGE).

Stage	Key settings
Models (S2)	224 px; AdamW, 20 epochs, batch 32, weighted CE; lr 10^{-3} (CNN, ScatNet), 10^{-4} (ResNet18); flips, $\pm 10^\circ$, shift, zoom, brightness
CV (S2)	5 grouped, stratified folds of the train split
XAI (S3)	IG 64 steps; Occlusion 16×16 , stride 8; LIME 1,000 samples, 16×16 patches
Joint (S4)	448 px; batch 16, half fractured; AdamW 10^{-4} , 20 epochs; $\lambda_d=1$, $\lambda_p=\lambda_a=\lambda_e=0.5$; warm-up 2 ep.; epoch by val AUC, threshold by val F1
YOLO (S5)	YOLOv8s (COCO), 640 px, 50 epochs, batch 16

ability along the way; lower = more faithful. A random patch order is the reference.

- **Hit rate** (pointing game): share of the 104 fractured test X-rays whose hottest point (after a small Gaussian smoothing) lies inside a radiologist’s box. A random point hits 1% (the boxes cover 1% of an image).

For detectors we report the hit rate of the most confident box (its centre inside a true box), IoU, and AP@0.5. All runs use seed 42 on a Kaggle T4 GPU (about 6 h for the whole notebook); Table II lists the settings.

IV. STAGE 2 — CNN VS SCATNET (+ RESNET18)

The three models differ only in the feature extractor (Fig. 2); they end with the *same* classifier (flatten $\rightarrow 512 \rightarrow 128 \rightarrow 2$, ReLU and dropout 0.5), whose input size alone changes.

- **CNN:** four learned blocks (conv, batch norm, ReLU, 2×2 max-pool) with 32, 64, 128 and 256 filters; the first filters are 7×7 . Every filter is *learned* from our 2,920 X-rays.
- **ScatNet:** a scattering transform with $J=4$ scales and $L=8$ angles. Its Morlet wavelets are *fixed*: it computes $|x * \psi_{j,\theta}| * \phi$ (first order) and $||x * \psi_1| * \psi_2| * \phi$ (second order),

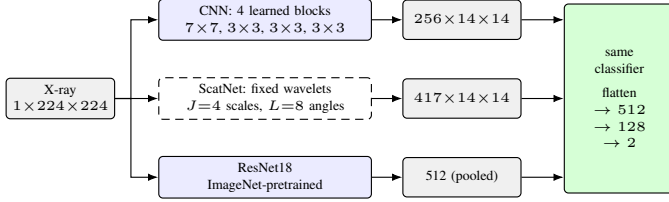


Fig. 2. **S2: three feature extractors, one classifier.** Blue: learned; dashed: fixed (nothing learned in ScatNet before the classifier). CNN and ScatNet both end on a 14×14 grid, so only the classifier’s input size differs (50,176 and 81,732 inputs; 26.1M and 41.9M parameters, almost all in the classifier). ResNet18 has 11.5M.

TABLE III

S2: CROSS-VALIDATION (MEAN $\pm$ STD OVER 5 FOLDS) AND TEST (586 X-RAYS, 104 FRACTURED). ALWAYS “NOT FRACTURED” SCORES 82.3% ACCURACY.

Model	5-fold CV		Test		
	Acc.	F1	Acc. [95% CI]	F1	Found
CNN	73.0 $\pm$ 9.0	35.7 $\pm$ 13.0	80.9 [77.8, 84.0]	23.3	16.3
ScatNet	77.5 $\pm$ 2.9	38.0 $\pm$ 2.6	80.0 [76.8, 83.3]	31.6	26.0
ResNet18	89.0$\pm$1.0	66.6$\pm$2.7	91.5 [89.2, 93.7]	73.7	67.3

417 maps of 14×14 , then log and batch norm. Nothing is learned before the classifier.

- **ResNet18** [5]: ImageNet-pretrained, fine-tuned; the grey X-ray is repeated on the three channels.

Training. Each model is first scored by 5-fold grouped cross-validation on the train split, then trained on the whole train split; the epoch with the best val accuracy is kept and the test split is used once.

Results and interpretation. Table III and the learning curves (Fig. 3) tell the same story. *CNN and ScatNet underfit*: their training loss stays near 0.6 and they find only 16–26% of the fractures; their test accuracy is *below* the 82.3% of always answering “not fractured”. They are statistically indistinguishable (McNemar: 25 vs 20 X-rays only one of them gets right, $p = 0.55$). Three reasons: (1) only 511 fractured X-rays to learn from; (2) a fracture is a thin line of a few pixels at 224 px; (3) flattening a 14×14 grid gives a classifier with 26–42M parameters, far too many for this data. ScatNet adds a fourth: its averaging makes it stable to small deformations, but a small break *is* a small deformation. *ResNet18 wins clearly* (91.5%, F1 73.7%; McNemar vs CNN 79 vs 17, vs ScatNet 86 vs 19, both $p < 0.001$): ImageNet pretraining gives it edge and texture detectors that our data alone cannot teach. It overfits after about 10 epochs (training loss falls, validation loss rises), which is why the best val epoch is kept. **Choice for the pipeline: ResNet18.**

Filters. Fig. 4 compares the first layers. The CNN’s 32 learned 7×7 filters are noisy blobs and edges that cover the frequency plane unevenly; ScatNet’s Morlet wavelets are clean oriented bands that cover every scale and orientation evenly by construction. Clean filters do not help here: what ScatNet lacks is not the first layer but a learned, local and selective

TABLE IV
THE SIX METHODS: WHAT THEY COMPUTE AND THEIR COST (\$ PER X-RAY, RESNET18 AT 224 PX, T4 GPU).

Method	Signal	Map	s/X-ray
Saliency	gradient	pixel	0.015
Integrated Gradients	path of gradients	pixel	0.194
Guided Backprop	guided gradient	pixel	0.010
Grad-CAM	gradient $\times$ features	layer grid	0.005
Occlusion	remove a window	window	0.767
LIME	remove patches, fit	patch	1.377

feature hierarchy above it.

V. STAGE 3 — SIX XAI METHODS

Each method gives a map $A(x)$ of the evidence for the fractured class (Table IV).

- **Saliency**: $|\partial z_c / \partial x|$, how much the score changes for a tiny change of each pixel.
- **Integrated Gradients**: the gradient averaged on the straight path from a black image x' to x , $A = (x - x') \odot \frac{1}{m} \sum_k \nabla z_c(x' + \frac{k}{m}(x - x'))$; the map sums to $z_c(x) - z_c(x')$.
- **Guided Backprop**: the gradient with negative signals stopped at every ReLU; sharp, but close to an edge detector.
- **Grad-CAM**: on the last convolutional layer $a \in \mathbb{R}^{K \times H \times W}$, $A = \text{ReLU}(\sum_k \alpha_k a_k)$ with α_k the mean gradient of z_c over channel k ; coarse but class-specific. It needs a learned convolutional layer, so it is *not applicable to ScatNet*.
- **Occlusion**: slide a black 16×16 window (stride 8) and record the drop of z_c .
- **LIME**: switch random sets of 16×16 patches off (1,000 samples) and fit a ridge regression from the patches to the score.

Occlusion from scratch. We re-implemented Occlusion in PyTorch: every window position becomes one black patch, all positions are scored in batches, and each pixel averages the drops of the windows that cover it. On the 10 explained test X-rays it equals Captum’s result for all three models (Pearson $r = 1.000$, largest difference $8.6 \cdot 10^{-6}$).

Results and interpretation. Fig. 5 shows the six methods on two fractured and two healthy test X-rays for each model; Table V gives the deletion test. Which method is best depends on the model. On the *CNN*, Grad-CAM is the most faithful (0.497): the last convolutional layer of a small CNN is where its decision is made. On *ScatNet* every method is close to random (0.552–0.588 vs 0.576): its evidence is spread over hundreds of wavelet coefficients, so no small region matters on its own. On *ResNet18* LIME is the most faithful (0.496 vs 0.700 for random): it measures exactly what the deletion test measures, the effect of removing patches. Saliency and Integrated Gradients are the least faithful there: gradients are local and saturate in a deep network. All maps are visibly

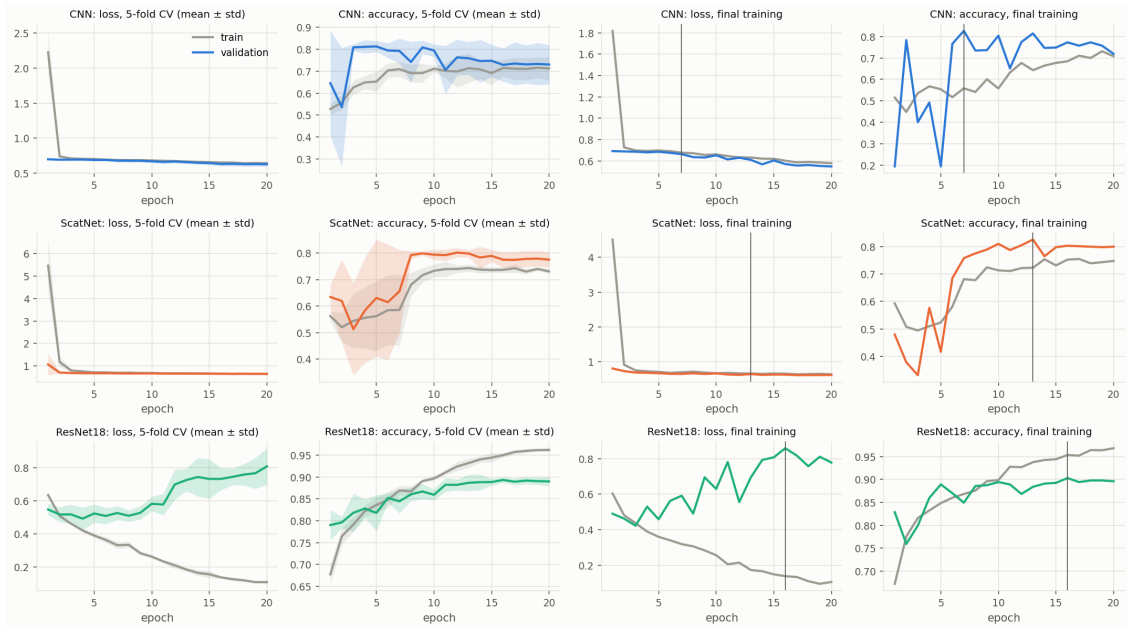


Fig. 3. **Learning curves**, train (grey) and validation (colour) in the same panels: 5-fold CV (mean $\pm$ std) and final training (vertical line: the epoch kept). CNN and ScatNet stay near loss 0.6 (underfitting); ResNet18 fits the training set and starts to overfit after about 10 epochs.

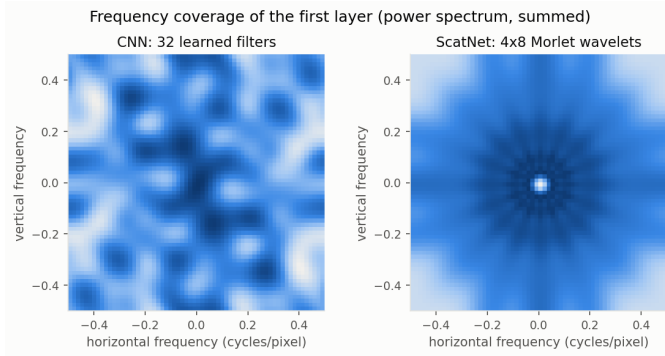


Fig. 4. **First-layer filters in frequency** (power spectrum summed over the filters). Left: the 32 learned CNN filters cover the plane unevenly. Right: ScatNet’s 4×8 fixed Morlet wavelets cover every scale and angle. The filters themselves are in `filters_cnn.png` and `filters_scatter.png`.

different (Fig. 5): the methods do not agree on one “true” explanation.

VI. STAGE 3 — FROM HEATMAP TO FRACTURE BOX

A heatmap becomes a box in three steps (Fig. 6): keep the positive evidence and smooth it (Gaussian, $\sigma = 4$ px); keep the pixels above 50% of the maximum; take the connected blob with the most evidence and draw its box. The hottest point gives the hit rate.

Results and interpretation. The maps rarely point at the fracture (Table VI). CNN and ScatNet stay at 0–4%, close to a random point: a model that barely detects fractures cannot point at them. ResNet18’s best method, Integrated Gradients, hits 23%: 23 times a random point, but still low. Four reasons: the boxes are tiny (1% of the image); at 224 px a hairline

TABLE V
S3: DELETION AUC (LOWER = MORE FAITHFUL) ON 10 TEST X-RAYS (5 PER CLASS). N/A: GRAD-CAM NEEDS A LEARNED CONVOLUTIONAL LAYER.

Method	CNN	ScatNet	ResNet18
Saliency	0.530	0.573	0.663
Integrated Gradients	0.510	0.572	0.637
Guided Backprop	0.511	0.552	0.555
Grad-CAM	0.497	n/a	0.580
Occlusion	0.572	0.588	0.562
LIME	0.528	0.554	0.496
Random order	0.570	0.576	0.700

TABLE VI
S3: HIT RATE (%) ON THE 104 FRACTURED TEST X-RAYS (224 PX). A RANDOM POINT HITS 1.0%.

Method	CNN	ScatNet	ResNet18
Saliency	3.8	0.0	3.8
Integrated Gradients	2.9	2.9	23.1
Guided Backprop	3.8	0.0	4.8
Grad-CAM	0.0	n/a	9.6
Occlusion	3.8	3.8	6.7
LIME	0.0	0.0	1.0

fracture is 1–2 pixels wide; a third of the fractures are missed by the model, so their “fractured” map has nothing to show; and the model also uses *shortcuts* such as metal plates and casts, which the faithful methods then honestly highlight.

A first attempt that failed. We first retrained ResNet18 with two extra losses on its 14×14 layer3: an Energy loss (share of the Grad-CAM heat outside the box [17]) and

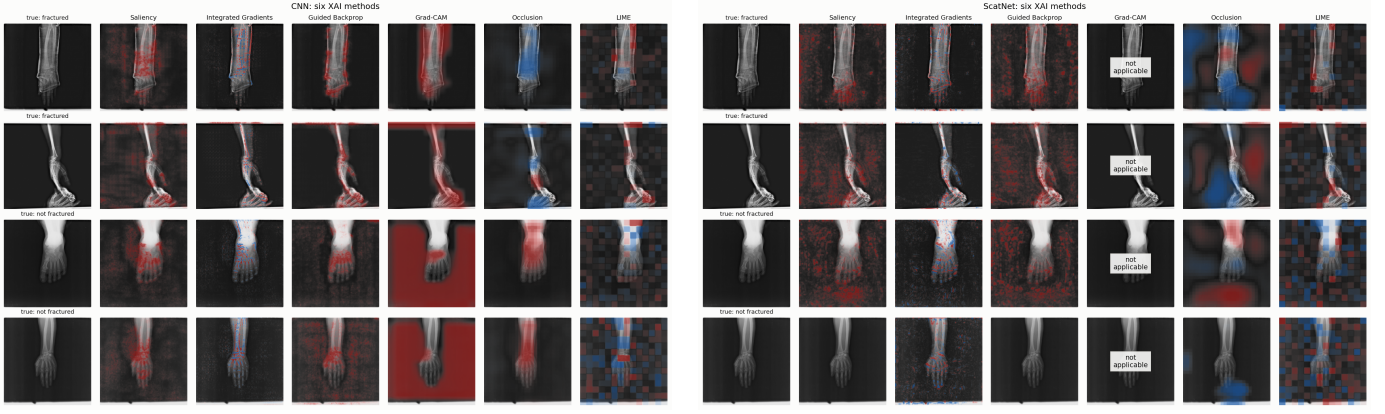


Fig. 5. **The six methods on the same test X-rays** (rows: 2 fractured, 2 healthy; red = evidence for the class, blue = against). Left: CNN; right: ScatNet (Grad-CAM not applicable). ResNet18 is in `xai_resnet18.png`.

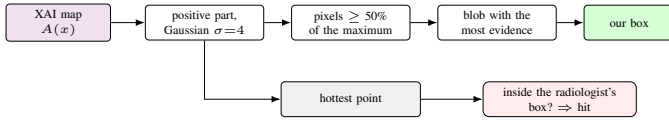


Fig. 6. **From an XAI map to a box.** The box is compared with the radiologist’s box (IoU); the hottest point gives the pointing-game hit rate.

a supervised contrastive loss (features inside the box alike, background apart). The Grad-CAM hit rate did not move (10.6%→10.6%, and 5.8% with the contrastive loss). The reasons shaped S4: Grad-CAM was *trained* on layer3 but *scored* on layer4; the ReLU inside Grad-CAM let the map collapse to zero, after which it gave no gradient; and the contrastive loss reached its lowest possible value at once (a box on bone and a background full of black air are trivially different), so it taught nothing.

VII. STAGE 4 — OUR METHOD: ONE JOINT MODEL

Idea. Instead of a classifier with a separate detector, one network classifies *and* finds the box, and its explanation is part of the loss (Fig. 7). The backbone is ResNet18 at 448 px (four times the pixels, so thin fractures survive); its layer3 gives a 28×28 grid a of 16-pixel cells shared by three heads:

- the **classifier**: layer4, average pooling and the same classifier as in S2;
- a **built-in detector** in the style of CenterNet [21]: a 3×3 and a 1×1 convolution give, per cell, a fracture-centre score, the centre offset and the box size;
- **Grad-CAM**, computed *during training* on the same grid, for the score $s = z_{\text{fractured}} - z_{\text{healthy}}$, without the final ReLU: $g = \sum_k \alpha_k a_k$.

The loss. With $p = \text{softmax}(\hat{g}/\tau)$ the Grad-CAM map turned into a probability over the cells ($\hat{g}$ standardised, $\tau = 0.5$), q the same for the detector’s centre map, B the cells of the (slightly grown) radiologist’s box, and $\tilde{x}$ the mirrored X-ray,

$$\mathcal{L} = \text{CE} + \lambda_d \mathcal{L}_{\text{det}} + r(t) [\lambda_p \mathcal{L}_{\text{point}} + \lambda_a \mathcal{L}_{\text{agree}} + \lambda_e \mathcal{L}_{\text{erase}}], \quad (1)$$

$$\mathcal{L}_{\text{point}} = -\log \sum_{i \in B} p_i, \quad (2)$$

$$\mathcal{L}_{\text{agree}} = \text{JS}(p, q) + \text{JS}(p, \text{flip}(p(\tilde{x}))), \quad (3)$$

$$\mathcal{L}_{\text{erase}} = \text{CE}(f(\text{blur}_B(x)), \text{healthy}) + \text{CE}(f(\text{blur}_{B'}(x)), \text{fractured}), \quad (4)$$

where $\mathcal{L}_{\text{det}}$ is CenterNet’s focal loss on the centre map plus an L1 loss on offset and size, JS the Jensen–Shannon divergence, blur_B a blur of the box region and B' the same box moved away from the fracture. The three explanation losses are applied to fractured X-rays (the mirror term to all) and ramp up after a 2-epoch warm-up $r(t)$.

Why each part should help.

- **Detection** (multi-task): to find the centre, the shared grid must encode *where* fractures are, so the classifier builds on localised features.
- **Pointing** is a smooth version of the hit rate itself. The softmax always has a gradient, so the map cannot collapse as in the failed attempt, and it is trained on the layer that is scored.
- **Agreement** is the contrastive (SimCLR-like [19]) part: two views of the same X-ray, the detector’s and the classifier’s, and the original and mirrored image [18], must point at the same place; its gradient updates *both* heads.
- **Erase** keeps the focus honest: a map can be moved without changing the reasoning [20], but the decision itself must change when the fracture is removed, and must not change when an equal patch elsewhere is removed. It is the training-time version of Occlusion [16].

Three versions use the same data, budget and seed: **A** (CE only; its detector is YOLO, S5), **B** (+ detector) and **C** (+ detector + explanation losses). Every batch is half fractured; the threshold on $p(\text{fractured})$ is chosen on val F1.

Results and interpretation. *Resolution and balanced batches* already help: A beats the 224-px ResNet18 (93.3% vs 91.5%, 80% of fractures found vs 67%) and its best map (Integrated Gradients) hits 54% instead of 23%. *The detector teaches the classifier where to look*: B raises every method

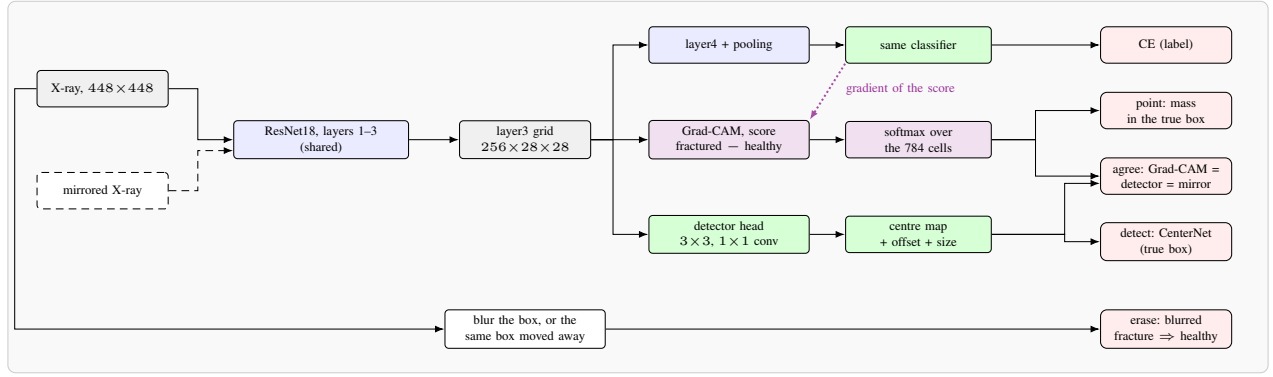


Fig. 7. **S4: the joint model.** One ResNet18 at 448 px; its layer3 grid feeds the classifier (through layer4), the built-in detector and Grad-CAM, computed during training (violet). The mirrored X-ray (dashed) is a second view for the agreement loss; the blurred X-rays test that the decision really comes from the fracture. A uses only the classification path, B adds the detector, C adds the three explanation losses.

TABLE VII

S4: CLASSIFICATION ON THE TEST SPLIT. RESNET18 IS THE S2 MODEL AT 224 PX; A, B, C AT 448 PX.

Model	Acc. [95% CI]	F1	Found	Spec.	AUC
ResNet18, 224	91.5 [89.2, 93.7]	73.7	67.3	96.7	91.2
A: separate	93.3 [91.3, 95.2]	81.0	79.8	96.3	94.4
B: + detector	94.2 [92.2, 96.1]	82.3	76.0	98.1	94.3
C: + XAI losses	93.0 [90.8, 94.9]	80.4	80.8	95.6	95.2

TABLE VIII

S4: HIT RATE (%) OF ALL SIX METHODS ON THE 104 FRACTURED TEST X-RAYS, AND THE EXACT McNEMAR TEST A VS C (SAME X-RAYS). ONLY GRAD-CAM IS PART OF C'S LOSS.

Method	A	B	C	A vs C (p)
Saliency	40.4	56.7	49.0	0.136
Integrated Gradients	53.8	59.6	61.5	0.077
Guided Backprop	32.7	50.0	51.9	0.0001
Grad-CAM	34.6	55.8	67.3	< 0.001
Occlusion	31.7	39.4	43.3	0.050
LIME	16.3	28.8	24.0	0.115
Mean of the six	34.9	48.4	49.5	

(mean hit rate 35%→48%, Grad-CAM 35%→56%), although no explanation is in its loss. *The explanation losses finish the job:* C's Grad-CAM hits 67% (70 vs 36 of the 104 X-rays, $p < 0.001$) and Guided Backprop improves significantly as well, while the other four methods improve without reaching significance in this run (Table VIII). Grad-CAM falls inside the detector's box for 87% of the X-rays in C, 62% in B and 33% in A (against YOLO).

The blur test (Table IX) shows that this is not cosmetic. Blurring the fracture does not change A's answer at all (80.6%→84.4%): A decides from the rest of the X-ray. B's confidence halves; C's falls to 2.5%, while blurring an equal patch elsewhere leaves it at 79%. C is trained with this blur, but on other X-rays: on the test split its decision depends on the fracture region. Faithfulness does not suffer: the mean deletion AUC is 0.55, 0.52 and 0.54 for A, B and C (random

TABLE IX

S4: BLUR TEST, MEAN $p(\text{FRACTURED})$ (%) OF THE FRACTURED TEST X-RAYS.

Model	As they are	Fracture blurred	Random patch blurred
A	80.6	84.4	80.9
B	77.0	47.7	77.7
C	83.7	2.5	79.4

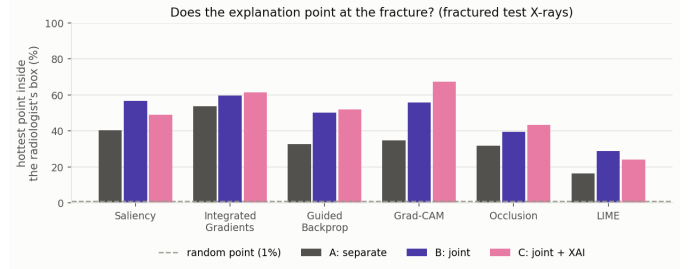


Fig. 8. **Where do the explanations point?** Hit rate of the six methods for A, B and C (fractured test X-rays).

0.70–0.74).

The classification metrics of A, B and C are within each other's confidence intervals (Table VII): our method changes *why* the model decides, not how often it is right. A second full run of the same code gave the same picture (accuracy 91.1, 93.3, 93.9% for A, B, C; Grad-CAM hit 30, 60, 65%; blur drop 2, 27, 81 points): the large effects repeat, the one-point differences in accuracy do not. The built-in detectors are weaker than YOLO (AP@0.5 31% for B and 22% for C, against 53%; Section VIII). **Choice for the pipeline: C**, also chosen by val AUC.

VIII. STAGE 5 — YOLO, “YOLO PROPOSES, C DECIDES” AND THE PIPELINE

YOLO. YOLOv8s (COCO-pretrained, 640 px, 50 epochs) is trained on the boxes of the fractured training X-rays. It reaches mAP@0.5 = 54.4% on the fractured test X-rays (the FracAtlas paper reports 56.2%), and its most confident box lands on the

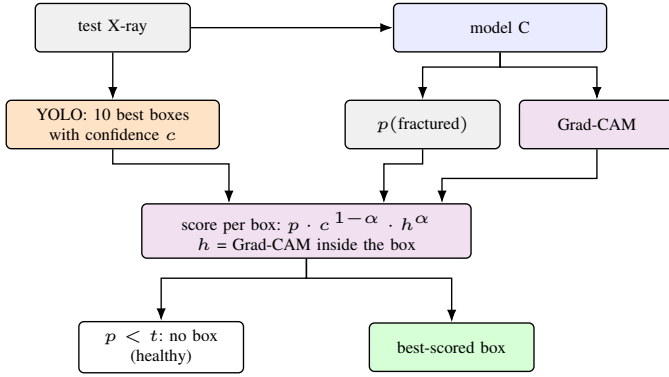


Fig. 9. **YOLO proposes, C decides.** The weight α of C’s Grad-CAM is chosen on the val X-rays ($\alpha = 0$ keeps YOLO’s own order); t is C’s decision threshold.

TABLE X

S5: BOX FINDERS ON THE SAME TEST X-RAYS. HIT RATE AND IOU ON THE 104 FRACTURED X-RAYS (MCNEMAR VS YOLO ALONE); AP@0.5 ON THE FRACTURED X-RAYS AND ON ALL 586 (BOXES ON HEALTHY X-RAYS ARE FALSE POSITIVES).

	Hit	IoU	p	AP frac.	AP all
YOLO alone	78.8	0.49	–	53.0	40.7
YOLO + C, re-ranked	76.0	0.47	0.375	49.7	46.9
+ healthy dropped	64.4	0.41	<0.001	46.6	44.6

fracture for 79% of them: better than any explanation (67%), because boxes are all it learns.

YOLO proposes, C decides (Fig. 9). YOLO’s 10 best boxes are re-scored with C’s confidence that the X-ray is fractured and with how much of C’s Grad-CAM falls inside each box. On val, $\alpha = 0.5$ hits 74.5% against 71.6% for YOLO’s own order. Optionally, X-rays that C calls healthy lose all their boxes.

Results and interpretation. The fusion does *not* place boxes better (Table X): 76% vs 79%, not a significant difference, and the val gain of 3 X-rays did not carry over to the test split. YOLO’s own ranking is already good, and C’s Grad-CAM, built on 16-pixel cells, is coarser than YOLO’s boxes. It does help where YOLO is weak: on all X-rays, AP@0.5 rises from 40.7% to 46.9%, because multiplying by $p(\text{fractured})$ pushes YOLO’s boxes on healthy X-rays to the bottom of the ranking. Dropping those boxes completely is worse (64% hits): C misses about one fracture in five, and their boxes are thrown away with them.

The pipeline. As in Linda [2], two opinions decide: C (“fractured?”) and YOLO (a confident box, $c \geq 0.25$). If they agree, the pipeline answers; if not, the X-ray goes to a radiologist’s review; the box shown is the one C chose among YOLO’s confident boxes. It decides 76% of the test X-rays with 96.6% accuracy, sends 24% to review, and misses 5.8% of the fractures (6 of 104), where both models say “healthy”. Fig. 10 shows why two opinions help: on two of C’s three surest missed fractures, YOLO draws its box on the fracture, so the pipeline asks for a review instead of missing them.

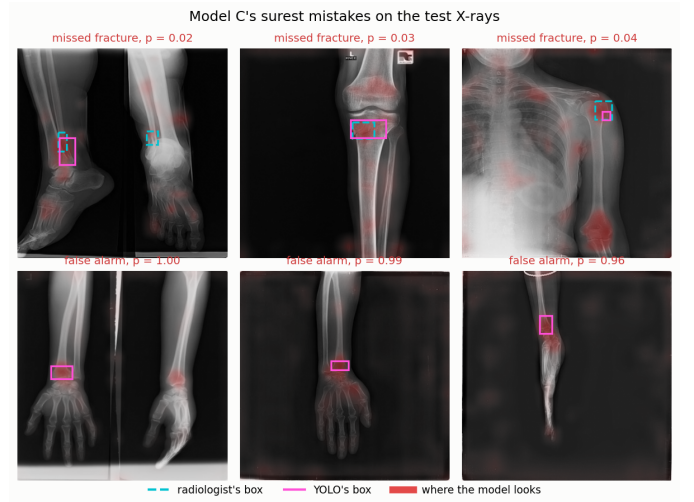


Fig. 10. **Model C’s surest mistakes.** Top: missed fractures (dashed: radiologist’s box; magenta: YOLO’s box; red: where C looks). Bottom: false alarms, mostly wrists where YOLO also draws a box.

IX. DISCUSSION AND LIMITATIONS

Four themes recur. First, *accuracy is not enough*: on a dataset with one positive in six, CNN and ScatNet reach 80% while finding a fifth of the fractures, and a model that is right can still look at plates and casts. Second, *data and resolution beat hand-crafted invariance*: fixed wavelets did not replace learned, pretrained features, and moving from 224 to 448 px helped every model and every map. Third, *an explanation becomes trustworthy only when the model is trained to look*: the faithful methods were honest from the start, but they showed shortcuts; training with a detector and with explanation losses moved the evidence onto the fracture, and the blur test confirms that the decision followed. Fourth, *a specialist stays the best at its job*: YOLO places boxes better than any explanation, and the best use of our classifier next to it is as a second opinion and a filter of false boxes.

Limitations. One dataset and one seed per run (two full runs of S4); 104 fractured test X-rays, so a difference of 4 points is 4 X-rays; the deletion test uses only 10 X-rays. Loss weights, thresholds and the fusion grid were set once, not tuned. The pointing game is strict (one point, a tiny box), and the boxes are rectangles around oblique fractures. C is trained with the same blur used in the blur test; the random-patch control and the five methods outside its loss are the independent evidence. The built-in detector remains far behind YOLO.

X. CONCLUSION

On FracAtlas, a CNN and a ScatNet with the same classifier are statistically tied and close to the trivial baseline, while a pretrained ResNet18 reaches 91.5%. Of six XAI methods, the most faithful depends on the model (Grad-CAM for the CNN, LIME for ResNet18), and our Occlusion matches Captum exactly; but the maps point at the fracture at most 23% of the time. Our joint model, a ResNet18 at 448 px that classifies, detects and is trained with explanation losses, keeps

the accuracy (93.0%) and makes the model look at the fracture: Grad-CAM hits 67% instead of 35%, and blurring the fracture now changes the answer (84%→3%). YOLO remains the best box finder (79%); our model helps it by discarding false boxes on healthy X-rays (AP@0.5 on all X-rays 40.7%→46.9%), and the two together decide 76% of the X-rays with 97% accuracy.

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